NANO 2314 Quiz 9 — after-quiz review

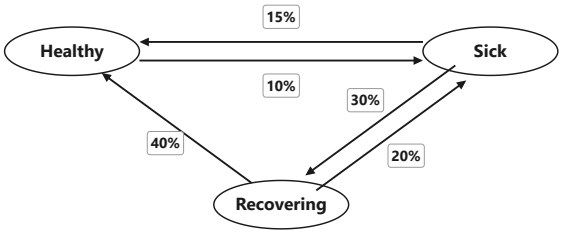
Quiz 9 was given on Friday, November 6, 2026. Everyone got the same booklet — there were no scrambled versions — so the answers below are the same for the whole class.

> Printable version, with every solution shown. Interactive page: https://tinyurl.com/N2314-F26-Quiz09-review.

Quiz 9 — November 6, 2026

1. (2 pts) A town of **1000 people** is split into three groups: **Healthy**, **Sick**, and **Recovering**. Every week, people move between the groups at the rates shown in the diagram. For example, the arrow from Healthy to Sick labeled 10% means that every week 10% of the healthy people become sick. The total population stays the same.

Let h_n, s_n, and r_n be the numbers of healthy, sick, and recovering people in week n. Write a **system of three equations** that gives h_n, s_n, and r_n in terms of the numbers in the week prior, h_{n-1}, s_{n-1}, and r_{n-1}.



The diagram as printed on the quiz.

Solution

each group this week = the people who **stay** + the people who **arrive**, both from last week

↓ Healthy: 90% stay; arrivals from Sick and Recovering

$$h_n = 0.90\,h_{n-1} + 0.15\,s_{n-1} + 0.40\,r_{n-1}$$

↓ the same for Sick and Recovering

$$s_n = 0.10\,h_{n-1} + 0.55\,s_{n-1} + 0.20\,r_{n-1}$$

$$r_n = 0\,h_{n-1} + 0.30\,s_{n-1} + 0.40\,r_{n-1}$$

> **Why:** Sick loses 15% + 30% = 45%, so 55% stay; Recovering loses 20% + 40% = 60%, so 40% stay. No arrow goes from Healthy to Recovering, so that term is 0.

> **In words.** Each new number is the people who stay in the group (100% minus everyone who leaves it) plus the people who arrive from the other groups (each arrow's rate times last week's number in the group the arrow starts from).

2. (2 pts) Rewrite your equations as one **eigenvalue equation**. It should have a 3×3 matrix, the eigenvalue λ, and the eigenvector $\begin{pmatrix} h \\ s \\ r \end{pmatrix}$ appearing on both sides of the equation.

Solution

the numbers on h_{n-1}, s_{n-1}, r_{n-1} in each equation become one row of the matrix

$$\begin{pmatrix} h_n \\ s_n \\ r_n \end{pmatrix} = \begin{pmatrix} 0.90 & 0.15 & 0.40 \\ 0.10 & 0.55 & 0.20 \\ 0 & 0.30 & 0.40 \end{pmatrix} \begin{pmatrix} h_{n-1} \\ s_{n-1} \\ r_{n-1} \end{pmatrix}$$

↓ drop subscripts; left side becomes λ times the vector

Conclusion

$$\begin{pmatrix} 0.90 & 0.15 & 0.40 \\ 0.10 & 0.55 & 0.20 \\ 0 & 0.30 & 0.40 \end{pmatrix} \begin{pmatrix} h \\ s \\ r \end{pmatrix} = \lambda \begin{pmatrix} h \\ s \\ r \end{pmatrix}$$

Why: an eigenvector comes out of the matrix as a multiple of itself, so the same vector, with no subscripts, appears on both sides; λ is that multiple.

In words. Put each equation's three rates in a row of the matrix, in the order h, s, r. For an eigenvector, multiplying by the matrix gives back the same vector times a number λ.

3. (2 pts) How many **eigenvalues** does the matrix in Problem 2 have? Out of those, why is $\lambda = 1$ the physically reasonable one for this town?

Solution

How many: 3. A 3×3 matrix has three eigenvalues (here 1, 0.6 and 0.25; you were not asked to find them).

Why λ = 1: the total population is conserved, so a population that has settled must stay the same from week to week: multiplying by the matrix gives back $1\times$ the same vector. Any other λ would scale the whole population every week — λ < 1 would shrink it and λ > 1 would grow it. (The eigenvectors for 0.6 and 0.25 also have negative entries, which cannot be numbers of people.)

4. (2 pts) Verify that the vector $(h, s, r) = (700,\ 200,\ 100)$ is an **eigenvector** with $\lambda = 1$, by multiplying your matrix from Problem 2 by this vector. Show the multiplication, and state what you conclude.

Solution

each **row** of the matrix times $(700, 200, 100)$, entry by entry, then add across the row

$$0.90(700) + 0.15(200) + 0.40(100) = 630 + 30 + 40 = 700$$

$$0.10(700) + 0.55(200) + 0.20(100) = 70 + 110 + 20 = 200$$

$$0(700) + 0.30(200) + 0.40(100) = 0 + 60 + 40 = 100$$

Why: add across each row, not down each column. Adding down the columns also gives 700, 200, 100 here, but only because each column adds up to 1; it is not matrix-times-vector multiplication.

Conclusion

$$\begin{pmatrix} 0.90 & 0.15 & 0.40 \\ 0.10 & 0.55 & 0.20 \\ 0 & 0.30 & 0.40 \end{pmatrix}\begin{pmatrix} 700 \\ 200 \\ 100 \end{pmatrix} = \begin{pmatrix} 700 \\ 200 \\ 100 \end{pmatrix} = 1 \cdot \begin{pmatrix} 700 \\ 200 \\ 100 \end{pmatrix}$$

an eigenvector with λ = 1

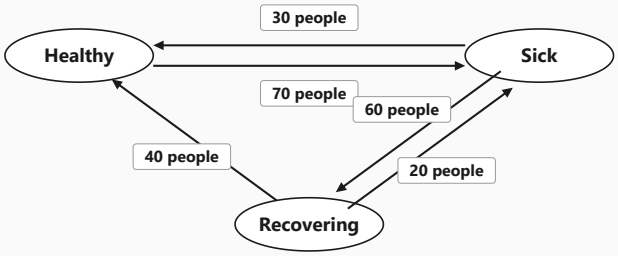
In words. Row by row, the matrix turns (700, 200, 100) back into (700, 200, 100). The result is 1 times the vector, so it is an eigenvector with eigenvalue 1.

5. (2 pts) What does the eigenvector $(700,\ 200,\ 100)$ **physically** tell you about the town as time goes on?

Solution

$\lambda = 1$: one week of moving leaves $(700, 200, 100)$ unchanged

↓ people keep moving, but the weekly flows balance



Healthy, 700 people: 70 leave, 30 + 40 = 70 arrive
Sick, 200 people: 30 + 60 = 90 leave, 70 + 20 = 90 arrive
Recovering, 100 people: 20 + 40 = 60 leave, 60 arrive

People moving each week at 700, 200, 100: into each group come exactly as many as leave it.

Conclusion

The town settles at 700 healthy, 200 sick and 100 recovering (70%, 20% and 10%), and those numbers stay the same week after week.

In words. After a long time, the town reaches an equilibrium of 700 healthy, 200 sick and 100 recovering people. People still move between the groups every week, but as many enter each group as leave it, so the three numbers no longer change.

EC. (1 pt) Extra credit. Show that the vectors $(0.6,\ 0.8)$ and $(-0.8,\ 0.6)$ are **normalized** and **orthogonal**.

Solution

length of $(0.6, 0.8)$:

$$\sqrt{0.6^2 + 0.8^2} = \sqrt{0.36 + 0.64} = 1$$

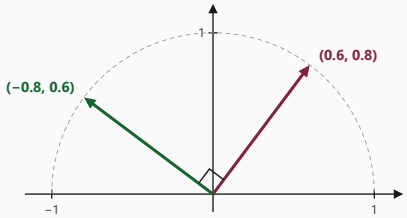
length of $(-0.8, 0.6)$:

$$\sqrt{(-0.8)^2 + 0.6^2} = \sqrt{0.64 + 0.36} = 1$$

↓ both lengths are 1: normalized

$$(0.6)(-0.8) + (0.8)(0.6) = -0.48 + 0.48 = 0$$

↓ the dot product is 0: orthogonal



Both arrows end on the circle of radius 1 and meet at a right angle.

> **In words.** Normalized means length 1: the square root of the sum of the squares is 1 for both vectors. Orthogonal means the dot product is 0: multiply matching entries and add, and you get 0.

Class results

13 booklets were graded.

- **Quiz score with the extra credit (out of 10, plus up to 1 extra credit):** average 9.29, median 10
- **Problem 1, the system of equations (out of 2):** average 1.90
- **Problem 2, the eigenvalue equation (out of 2):** average 1.85
- **Problem 3, how many eigenvalues, and why λ = 1 (out of 2):** average 1.77
- **Problem 4, checking the eigenvector by multiplying (out of 2):** average 1.69
- **Problem 5, what the eigenvector means (out of 2):** average 1.75
- **Extra credit (out of 1):** average 0.33

Where the class had the most trouble

These cost the most points. If you lost points here, start your review with them.

1. **Extra credit — what "normalized" and "orthogonal" mean.** Only 2 of 13 booklets earned the full point. Reasons such as "they point in opposite directions", "they have the same magnitude" or "the signs

are reversed" do not show either property. Normalized means **length 1**: take the square root of the sum of the squares. Orthogonal means **dot product 0**: multiply matching entries and add.

2. **Problem 4 — multiply across the rows.** Three booklets added the products down each column of the matrix instead of across each row. That happens to give 700, 200 and 100 back, but only because every column adds up to 1; it is not how a matrix multiplies a vector. Two more booklets set the rows up correctly and slipped in the arithmetic. Each row times the vector, then add.
3. **Problem 5 — say what settles, with the numbers.** Three booklets said only that the total stays at 1000 (given in the problem), or named just one group, or only compared the groups. The eigenvector says the town settles at 700 healthy, 200 sick and 100 recovering, and stays there week after week.
4. **Problem 2 — the same vector on both sides.** Two booklets kept subscripts on one side (h_n, s_n, r_n) and one put λ in twice (in the matrix and in front of the vector). In an eigenvalue equation the same vector, with no subscripts, appears on both sides, and λ appears once.
5. **Problem 3 — the count.** Two booklets wrote 1 or 2 eigenvalues. A 3×3 matrix has three.

Where to review each idea

Homework and earlier quizzes to revisit, problem by problem

- **Problem 1 (the system of equations):** the *Eigen Solver* homework, Problem 2(a)–(b) (three teams), and Quiz 8, Problem 2.
- **Problem 2 (the eigenvalue equation):** the *Eigen Solver* homework, Problem 2(c)–(d), and the *Eigenvectors* homework, Problem 2(b).
- **Problem 3 (how many eigenvalues; λ = 1):** the *Eigenvectors* homework, Problem 2(c) (the eigenvalue from conservation), and the *Eigen Solver* homework, Problem 3 (the solver prints three eigenvalues for a 3×3 matrix).
- **Problem 4 (matrix times vector):** the matrix–vector worksheet of October 28, and the *Eigen Solver* homework, Problem 2(c).
- **Problem 5 (what the eigenvector means):** the *Eigenvectors* homework, Problem 2(d), and the *Eigen Solver* homework, Problem 3(c).
- **Extra credit (normalized and orthogonal):** the November 4 notes, and the *Linear Algebra in Perspective* homework, Problems 7–8.
- **All of it, with fresh numbers:** the Quiz 9 Practice Drill and the Quiz 9 review packet.

Still stuck? Bring it to office hours.